陈宪章

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个人信息

出生年月：1993.09　　性别：男　　政治面貌：中共党员

工作经历

Ningbo Institute of Digital Twin / 宁波东方理工大学　2025.12–至今
Research Assistant Professor；周通课题组。

东方理工高等研究院 / 宁波东方理工大学　2023.09–2025.11
博士后，凝聚态理论；合作导师：周通。

教育背景

兰州大学，博士　2016.09–2023.06
理论物理；导师：黄亮教授。

斯特拉斯堡大学，联合培养博士　2020.12–2022.11
IPCMS，凝聚态理论；导师：Rodolfo A. Jalabert 教授。

兰州大学，本科　2012–2016
理论物理。

研究方向

研究量子输运与拓扑量子器件，关注 Majorana 模、磁性织构与交错磁体。结合解析模型与数值模拟，研究微观电子结构、拓扑性质和可测量输运信号之间的联系。此前的研究包括石墨烯纳米结构中的扫描栅显微技术、量子疤痕及多体能谱统计。

奖项荣誉

- 入选欧洲博士学院（Collège doctoral Européen）国际博士生项目（PDI），2021 年。
- 中国留学基金委奖学金（CSC），2020 年。
- 第五届全国统计物理大会优秀海报奖，2019 年。
- 硕士生国家奖学金，2016 年。

发表论文与预印本

已发表论文

[9] Ziye Zhu, Richang Huang, **X.-Z. Chen**, Xunkai Duan, Jiayong Zhang, Igor Zutic, Tong Zhou, *Altermagnetic Proximity Effect*, Phys. Rev. Lett. 136, 186702 (2026).

[8] Zhou Cui, Ziye Zhu, Xunkai Duan, Bowen Hao, **X.-Z. Chen**, Jiayong Zhang, Tong Zhou, *Symmetry-Driven Unconventional Magnetoelectric Coupling in Perovskite Altermagnets: From Bulk to the Two-Dimensional Limit*, Advanced Science, e00004 (2026).

[7] Sihui Wu, Jiahui Qian, Xunkai Duan, **X.-Z. Chen**, Zhou Cui, Jiayong Zhang, Chunlan Ma, Tong Zhou, *Electrically switchable quantum anomalous and spin Hall states in two-dimensional honeycomb lattices with non-Dirac bands*, Phys. Rev. B 111, 235414 (2025).

[6] **X.-Z. Chen**, Guillaume Weick, Dietmar Weinmann, Rodolfo A. Jalabert, *Scanning gate microscopy in graphene nanostructures*, Phys. Rev. B 107, 085420 (2023).

[5] **X.-Z. Chen**, Zhen-Qi Chen, Liang Huang, Celso Grebogi, Ying-Cheng Lai, *Many-body spectral statistics of relativistic quantum billiard systems*, Phys. Rev. Research 5, 013050 (2023).

[4] Xiao-Liang Li, **X.-Z. Chen**, Chen-Rong Liu, Liang Huang, *Quantization condition of scarring states in complex soft-wall quantum billiards*, Acta Phys. Sin. 69, 080506 (2020).

[3] Guan-Qun Zhang, **X.-Z. Chen**, Li Lin, Hai-Lin Peng, Zhong-Fan Liu, Liang Huang, Ning Kang, Hong-Qi Xu, *Transport signatures of relativistic quantum scars in a graphene cavity*, Phys. Rev. B 101, 085404 (2020).

[2] Chen-Rong Liu, Pei Yu, **X.-Z. Chen**, Hong-Ya Xu, Liang Huang, Ying-Cheng Lai, *Enhancing von Neumann entropy by chaos in spin-orbit entanglement*, Chin. Phys. B 28, 100501 (2019).

[1] Chen-Rong Liu, **X.-Z. Chen**, Hong-Ya Xu, Liang Huang, Ying-Cheng Lai, *Effect of chaos on two-dimensional spin transport*, Phys. Rev. B 98, 115305 (2018).

预印本

[P3] **X.-Z. Chen**, Jiayong Zhang, Bowen Hao, Jiahui Qian, Ziye Zhu, Igor Zutic, Zhenyu Zhang, Tong Zhou, *Altermagnets Enable Gate-Switchable Helical and Chiral Topological Transport with Spin-Valley-Momentum-Locked Dual Protection*, arXiv:2603.06487.

[P2] Ziye Zhu, **X.-Z. Chen**, Xunkai Duan, Zhou Cui, Jiayong Zhang, Igor Zutic, Tong Zhou, *Altermagnetoelectric Spin Field Effect Transistor*, arXiv:2512.02974.

[P1] Richang Huang, Yongliang Hu, **X.-Z. Chen**, Peng Yu, Igor Zutic, Tong Zhou, *Towards Scalable Braiding: Topological Superconductivity Unlocked under Nearly Arbitrary Magnetic Field Directions in Planar Josephson Junctions*, arXiv:2504.20031.

会议展示

报告

- Novel quantum spin and quantum anomalous Hall effects based on altermagnet.
  CPS Meeting 2025, Harbin, China, 09/2025.
- Many-body spectral statistics of relativistic quantum billiards systems.
  DPG Meeting 2022, Regensburg, Germany, 09/2022.

海报

- Transport signatures of scars in graphene dots at low magnetic field.
  GDR 2426, Aussois, France, 12/2021.
- Many-body spectral statistics of relativistic quantum billiard systems.
  The 5th National Statistical Physics Conference (SPCSC), Hefei, China, 07/2019.
- Transport signatures of relativistic quantum scars in a graphene cavity.
  International Workshop on Experimental and Theoretical Developments on Complex Quantum Systems, Lanzhou, China, 07/2018.

参与项目

- Spectral statistics of a class of pseudo-integrable quantum billiards.
  NSFC，项目编号 12175090，2021 年，经费 63 万元，参与。
- Study of spin anomalous effects in Dirac billiards.
  NSFC，项目编号 11775101，2017 年，经费 58 万元，参与。